Pham Hung Quoc Viet

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OBJECTIVE

3rd-year student at University of Information Technology – VNU-HCM (UIT) with a solid foundation in Probability & Statistics, Time-Series Analysis, and Machine Learning. Experienced in building high-throughput stream processing pipelines, engineering rolling-window statistical features, and benchmarking predictive models using Python and PySpark. Seeking a **Quantitative Trading / Fintech Intern** position to apply mathematical, statistical, and algorithmic modeling to develop, backtest, and optimize data-driven trading strategies.

EDUCATION

University of Information Technology – VNU-HCM (UIT) — 2023 – 2027 (expected)

Bachelor of Information Systems — GPA: 8.06/10

Relevant Coursework: Probability & Statistics, Applied Statistics, Machine Learning, Data Mining, Database Management Systems

SKILLS

Programming	Python, C++, SQL, Java, PHP
Quantitative	Probability & Statistics, Time-Series Analysis, Statistical Modeling
Machine Learning	Scikit-learn, XGBoost, LightGBM, TensorFlow, Optuna
Data & Streaming	Pandas, NumPy, PySpark, Kafka, PostgreSQL
Tools & Languages	Git/GitHub, Docker, Jupyter Notebook, IELTS 6.5 (2022)

PROJECTS

NYC Taxi — Real-time Fleet Intelligence & Anomaly Detection — 2026

PySpark, Kafka, Docker, PostgreSQL, Streamlit, Pandas, Scikit-learn | GitHub

- **High-Frequency Data Ingestion:** Developed a low-latency Kafka pipeline to ingest high-frequency event streams at 200 msg/s, simulating real-time market data architectures.
- **Time-Series Feature Engineering:** Leveraged PySpark Structured Streaming to compute 14 rolling-window statistical features (e.g., moving averages, volatility metrics) on live data streams.
- **Anomaly & Regime Shift Detection:** Designed an Isolation Forest and Minimum Covariance Determinant (MinCovDet) ensemble with strict voting to identify statistical anomalies and regime shifts.
- **Automated Model Retraining:** Managed continuous data ingestion to PostgreSQL and implemented automated model evaluation and parameter retraining triggered every 25 streaming batches.

Bank Customer Churn Prediction & Risk Attribution — 2026

Python, Scikit-learn, TensorFlow, XGBoost, LightGBM, SMOTE, SHAP, Optuna, Pandas | GitHub

- **Statistical Preprocessing & EDA:** Conducted extensive exploratory data analysis; applied Winsorization to mitigate outliers, improving the signal-to-noise ratio in predictive datasets.
- **Class Balancing & Data Augmentation:** Implemented SMOTE to resolve class imbalance (20.4% positive rate), statistically augmenting data from 6,400 to 10,192 samples to enhance model sensitivity.
- **Algorithmic Benchmarking:** Trained, optimized, and evaluated 7+ predictive models (XGBoost, LightGBM, Random Forest) utilizing Optuna and GridSearchCV for rigorous hyperparameter tuning.
- **Risk Attribution & Explainability:** Utilized SHAP and LIME to dissect feature contributions; employed Jaccard similarity analysis to quantify the stability of key risk variables across subsets.